

(数值分析典型例题) 差分格式截断误差的首项的求法和绝对稳定区间

定义(绝对稳定区间). 在使用差分格式解微分方程时, 任何一步产生的舍入误差在以后的计算引起的误差逐渐减弱, 则称差分格式关于 $z := \lambda h (\lambda < 0)$ 是**绝对稳定**的, 使差分格式稳定的所有 $z = \lambda h$ 的集合称为**绝对稳定区间**. 根据定义, 绝对稳定区间总是 $(-\infty, 0)$ 的子集.

我们介绍一个很有用的引理, 无论是在迭代法还是求稳定区间这里都很有用:

引理 若 b 和 c 都是实数, 则二次方程

$$x^2 - bx + c = 0 \quad (1)$$

的根的模小于 1 的充分必要条件是

$$|c| < 1, \quad |b| < 1 + c. \quad (2)$$

证明 先设方程(1)的根 x_1, x_2 为复数, 此时不妨设

$$x_1 = \rho(\cos \theta + \sqrt{-1} \sin \theta), \quad x_2 = \rho(\cos \theta - \sqrt{-1} \sin \theta),$$

其中 ρ 为实数, 从而有

$$x_1 + x_2 = 2\rho \cos \theta = b, \quad x_1 x_2 = \rho^2 = c > 0.$$

若 $|x_1| < 1, |x_2| < 1$, 则

$$|c| = |x_1||x_2| < 1, \quad |b| = |2\rho \cos \theta| \leq 2|\rho| < 1 + \rho^2 = 1 + c.$$

反之, 若 $|c| < 1$, 则

$$|x_1| = |x_2| = |\rho| = \sqrt{c} < 1.$$

其次, 设方程(1)的根为实数. 若 $|x_1| < 1, |x_2| < 1$, 则 $|c| = |x_1||x_2| < 1$.

由于 $b = x_1 + x_2$, 因此, 若 $x_1 + x_2 \geq 0$, 则

$$1 + c - |b| = 1 + x_1 x_2 - (x_1 + x_2) = (1 - x_1)(1 - x_2) > 0;$$

若 $x_1 + x_2 < 0$, 则

$$1 + c - |b| = 1 + x_1x_2 + x_1 + x_2 = (1 + x_1)(1 + x_2) > 0,$$

因此,无论在何种情形下,都有 $|b| < 1 + c$.

反之,若(2)式成立,则有

$$|c| = |x_1||x_2| < 1, \quad |x_1x_2| < 1, \quad |x_1 + x_2| < 1 + x_1x_2,$$

即 x_1, x_2 同时满足不等式 $|x_1x_2| < 1, (x_1 - 1)(x_2 - 1) > 0$ 和 $(x_1 + 1)(x_2 + 1) > 0$.
从而推得 $-1 < x_1 < 1, -1 < x_2 < 1$, 即 $|x_1| < 1, |x_2| < 1$. $\square$

例题

例1. $y_{n+1} = y_n + \frac{h}{3}(2f_n + f_{n+1})$.

解. 截断误差为

$$T = y(x_{n+1}) - y(x_n) - \frac{h}{3}(2y'(x_n) + y'(x_{n+1})).$$

考虑在 x_n 处的Taylor展开式:

$$\begin{aligned} y(x_{n+1}) &= y(x_n + h) \\ &= y(x_n) + hy'(x_n) + \frac{h^2}{2}y''(x_n) + \mathcal{O}(h^3); \\ y'(x_{n+1}) &= y'(x_n + h) \\ &= y'(x_n) + hy''(x_n) + \mathcal{O}(h^2); \\ \frac{h}{3}(2y'(x_n) + y'(x_{n+1})) &= hy'(x_n) + \frac{h^2}{3}y''(x_n) + \mathcal{O}(h^3); \\ T &= \frac{h^2}{6}y''(x_n) + \mathcal{O}(h^3), \end{aligned}$$

故局部截断误差首项为 $\frac{h^2}{6}y''(x_n)$, 具有1阶精度.

绝对稳定区间: 考虑 $f(x, y) = y' = \lambda y (\lambda < 0)$, 代入得到

$$\begin{aligned} y_{n+1} &= y_n + \frac{h}{3}(2\lambda y_n + \lambda y_{n+1}) \\ &= y_n + \frac{z}{3}(2y_n + y_{n+1}) \quad (z := h\lambda) \\ \implies y_{n+1} \left(1 - \frac{z}{3}\right) &= y_n \left(1 + \frac{2}{3}z\right) \\ R(z) &:= \frac{1 + \frac{2}{3}z}{1 - \frac{1}{3}z} = \frac{3 + 2z}{3 - z}. \end{aligned}$$

令 $|R(z)| < 1$ 得到 $|3 + 2z|^2 < |3 - z|^2$, 令 $z = x + \sqrt{-1}y (x, y \in \mathbb{R})$, 整理得到 $(x + 3)^2 + y^2 < 0$, 投影到 x 轴即得绝对稳定区间为 $(-6, 0)$. $\square$

例2(课本206页5(2)). $y_{n+1} = y_n + \frac{h}{2}(3f_n - f_{n-1})$.

解. 截断误差为

$$T = y(x_{n+1}) - y(x_n) - \frac{h}{2}(3y'(x_n) - y'(x_{n-1})).$$

考虑在 x_n 处的Taylor展开式:

$$\begin{aligned}y(x_{n+1}) &= y(x_n + h) \\&= y(x_n) + hy'(x_n) + \frac{h^2}{2}y''(x_n) + \frac{h^3}{6}y'''(x_n) + \mathcal{O}(h^4); \\y'(x_{n+1}) &= y'(x_n + h) \\&= y'(x_n) + hy''(x_n) + \frac{h^2}{2}y'''(x_n) + \mathcal{O}(h^3); \\\frac{h}{2}(3y'(x_n) - y'(x_{n-1})) &= hy'(x_n) + \frac{h^2}{2}y''(x_n) - \frac{h^3}{4}y'''(x_n) + \mathcal{O}(h^4); \\T &= \frac{1}{6}h^3y'''(x_n) + \frac{1}{4}h^3y'''(x_n) + \mathcal{O}(h^4) \\&= \frac{5h^2}{12}y'''(x_n) + \mathcal{O}(h^4),\end{aligned}$$

故局部截断误差首项为 $\frac{5h^2}{12}y'''(x_n)$, 具有2阶精度.

绝对稳定区间: 考虑 $f(x, y) = y' = \lambda y (\lambda < 0)$, 代入得到

$$\begin{aligned}y_{n+1} &= y_n + \frac{h}{2}(3\lambda y_n - \lambda y_{n-1}) \\&\iff y_{n+1} - \left(1 + \frac{3}{2}z\right)y_n + \frac{1}{2}zy_{n-1} = 0.\end{aligned}$$

因此即求 z 的范围, 使得 $r^2 - (1 + 3/2z)r + \frac{1}{2}z = 0$ 的根的模长小于1. 直接由**引理**有

$$\begin{cases} |1 + \frac{3}{2}z| < \frac{1}{2}z + 1, \\ |\frac{1}{2}z| < 1, \\ z < 0, \end{cases}$$

解得 $z \in (-1, 0)$. $\square$

 如果考试时不用这个引理呢? 那就两种选择.

Option 1. 直接讨论二次函数就好了. 注意判别式 $\Delta = \frac{1}{4}z^2 + \frac{1}{2}z + 1 > 0$, 因此该二次函数 $f(r) := r^2 - (1 + \frac{3}{2}z)r + \frac{1}{2}z$ 具有两个互异实根, 容易知道两实根的绝对

值小于1的充要条件是端点值大于0, 且对称轴在 $(-1, 1)$ 内, 即

$$f(-1) > 0, f(1) > 0, -1 < \frac{1 + \frac{3}{2}z}{2} < 1,$$

解得 $z \in (-1, 0)$.

Option 2. 证明引理然后使用.

例3(课本206页5(3)).
$$\begin{cases} \bar{y}_{n+1} = y_n + \frac{3}{4}hf(x_n, y_n), \\ y_{n+1} = y_n + \frac{h}{3}[f(x_n, y_n) + 2f(x_n + \frac{3}{4}h, \bar{y}_{n+1})]. \end{cases}$$

解. 先写成一个完整的迭代格式:

$$y_{n+1} = y_n + \frac{h}{3}[f(x_n, y_n) + 2f(x_n + \frac{3}{4}h, y_n + \frac{3}{4}hf(x_n, y_n))].$$

截断误差为

$$T = y(x_{n+1}) - y_n(x_n) - \frac{h}{3} \left(f(x_n, y_n) + 2f \left(x_n + \frac{3}{4}h, y_n + \frac{3}{4}hf(x_n, y_n) \right) \right).$$

考虑在 x_n 处的Taylor展开式:

$$\begin{aligned} y(x_{n+1}) &= y(x_n + h) \\ &= y(x_n) + hy'(x_n) + \frac{h^2}{2}y''(x_n) + \frac{h^3}{6}y'''(x_n) + \mathcal{O}(h^4); \\ f \left(x_n + \frac{3}{4}h, y_n + \frac{3}{4}hf(x_n, y_n) \right) \\ &= f(x_n, y_n) + \frac{3}{4}hf_x(x_n, y_n) + \frac{3}{4}hf_y(x_n, y_n)f(x_n, y_n) \\ &\quad + \frac{1}{2} \left[f_{xx}(x_n, y_n) \frac{9}{16}h^3 + 2f_{xy}(x_n, y_n) \frac{9}{16}h^3 f(x_n, y_n) + f_{yy}(x_n, y_n) \frac{9}{16}h^3 \right] \\ &\quad + \mathcal{O}(h^3) \\ &= y'(x_n) + \frac{3}{4}hy''(x_n) + \frac{9}{32}h^2[y'''(x_n) - f_y(x_n, y_n)y''(x_n)] + \mathcal{O}(h^3); \\ T &= \left[-\frac{1}{48}y'''(x_n) + \frac{3}{16}f_y(x_n, y_n)y'(x_n) \right] h^3 + \mathcal{O}(h^4). \end{aligned}$$

故局部截断误差首项为 $\left[-\frac{1}{48}y'''(x_n) + \frac{3}{16}f_y(x_n, y_n)y'(x_n) \right] h^3$, 具有2阶精度.

下面求绝对稳定区间. 考虑 $f(x, y) = y' = \lambda y (\lambda < 0)$, 代入得

$$\begin{aligned} y_{n+1} &= y_n + \frac{h}{3} [\lambda y_n + 2(1 + \frac{3}{4}h\lambda)y_n] \\ &= y_n + \frac{z}{3} \left[y_n + 2 \left(1 + \frac{3}{4}z \right) y_n \right] \quad (z := h\lambda) \\ &= (1 + z + \frac{z^2}{2})y_n, \\ R(z) &:= 1 + z + \frac{z^2}{2}. \end{aligned}$$

令 $|R(z)| < 1$, 只需考虑 $z \in \mathbb{R}_{<0}$, 注意到

$$\left| 1 + z + \frac{z^2}{2} \right| = \left| \frac{1}{2}(z+1)^2 + \frac{1}{2} \right| = \frac{1}{2}(z+1)^2 + \frac{1}{2} \geq 0,$$

故绝对稳定区间为 $\{z \in \mathbb{R}_{<0} : (z+1)^2 < 1\} = (-2, 0)$. $\square$

例4(课本206页5(4)). $y_{n+1} = y_{n-1} + \frac{h}{3}(f_{n-1} + 4f_n + f_{n+1})$.

解. 局部截断误差为

$$T_{n+1} = y(x_{n+1}) - y(x_{n-1}) - \frac{h}{3} [y'(x_{n-1}) + 4y'(x_n) + y'(x_{n+1})].$$

在 x_n 处进行Taylor展开得到

$$\begin{aligned} y(x_{n+1}) &= y_n + hy'_n + \frac{h^2}{2}y''_n + \frac{h^3}{6}y'''_n + \frac{h^4}{24}y_n^{(4)} + \frac{h^5}{120}y_n^{(5)} + \mathcal{O}(h^6), \\ y(x_{n-1}) &= y_n - hy'_n + \frac{h^2}{2}y''_n - \frac{h^3}{6}y'''_n + \frac{h^4}{24}y_n^{(4)} - \frac{h^5}{120}y_n^{(5)} + \mathcal{O}(h^6). \end{aligned}$$

相减得:

$$\begin{aligned} y(x_{n+1}) - y(x_{n-1}) &= 2hy'_n + \frac{2h^3}{6}y'''_n + \frac{2h^5}{120}y_n^{(5)} + \mathcal{O}(h^7) \\ &= 2hy'_n + \frac{1}{3}h^3y'''_n + \frac{1}{60}h^5y_n^{(5)} + \mathcal{O}(h^7). \end{aligned}$$

再展开后面的:

$$\begin{aligned} y'(x_{n+1}) &= y'_n + hy''_n + \frac{h^2}{2}y'''_n + \frac{h^3}{6}y_n^{(4)} + \frac{h^4}{24}y_n^{(5)} + \mathcal{O}(h^5) \\ y'(x_{n-1}) &= y'_n - hy''_n + \frac{h^2}{2}y'''_n - \frac{h^3}{6}y_n^{(4)} + \frac{h^4}{24}y_n^{(5)} + \mathcal{O}(h^5). \end{aligned}$$

相加并加上中间项 $4y'_n$:

$$\begin{aligned} S &= (y'_n - hy''_n + \cdots) + 4y'_n + (y'_n + hy''_n + \cdots) \\ &= 6y'_n + h^2 y'''_n + \frac{2h^4}{24} y_n^{(5)} + \mathcal{O}(h^6) \\ &= 6y'_n + h^2 y'''_n + \frac{1}{12} h^4 y_n^{(5)} + \mathcal{O}(h^6). \end{aligned}$$

故局部截断误差为

$$\begin{aligned} T &= \left(2hy'_n + \frac{1}{3}h^3 y'''_n + \frac{1}{60}h^5 y_n^{(5)} \right) - \left(2hy'_n + \frac{1}{3}h^3 y'''_n + \frac{1}{36}h^5 y_n^{(5)} \right) + \mathcal{O}(h^7) \\ &= -\frac{1}{90}h^5 y^{(5)}(x_n) + \mathcal{O}(h^7), \end{aligned}$$

故**局部截断误差的首项**为: $-\frac{1}{90}h^5 y^{(5)}(x_n)$, 具有 **4 阶** 精度. 代入差分格式:

$$\begin{aligned} y_{n+1} &= y_{n-1} + \frac{z}{3}(y_{n-1} + 4y_n + y_{n+1}) \\ \iff \left(1 - \frac{z}{3}\right)y_{n+1} - \frac{4z}{3}y_n - \left(1 + \frac{z}{3}\right)y_{n-1} &= 0. \end{aligned}$$

对应的特征方程为:

$$f(r) := (3 - z)r^2 - 4zr - (3 + z) = 0.$$

下面确定 z 的范围使 $|r| < 1$. 首先

$$\Delta = (-4z)^2 - 4(3 - z)(-(3 + z)) = 16z^2 + 4(9 - z^2) = 12z^2 + 36 > 0,$$

因此**两根恒为相异实根**. 由Vieta定理,

$$|r_1 r_2| = \left| \frac{-(3 + z)}{3 - z} \right| < 1,$$

解得 $z < 0$. 此时首项系数 $a = 3 - z > 0$, 抛物线开口向上. 两根在 $(-1, 1)$ 内需满足:

$$f(1) > 0, \quad f(-1) > 0, \quad -1 < \frac{-b}{2a} < 1$$

$f(1) > 0$ 与 $f(-1) > 0$ 矛盾, 故绝对稳定区间为 $\emptyset$. $\square$

课本206页5的(5)阶数比较高, 和前面思路完全一样, 无非就是多展开几项, 我们就留作练习了. 下面来看一个最为麻烦的, 需要多次展开.

✎ 隐式公式所用到的二元Taylor展开式:

$$f(x_n + \Delta x, y_n + \Delta y) = f + f_x \Delta x + f_y \Delta y + \frac{1}{2} (f_{xx} (\Delta x)^2 + 2f_{xy} \Delta x \Delta y + f_{yy} (\Delta y)^2) + \dots$$

以及全微分公式

$$y'' = f_x + f f_y, \quad y''' = f_{xx} + 2f f_{xy} + f^2 f_{yy} + f_y y''.$$

例5. 已知求解初值问题

$$\begin{cases} y' = f(x, y), \\ y(a) = y_0, \end{cases} \quad x \in [a, b]$$

的数值解公式为

$$y_{n+1} = y_n + hf \left(x_n + \frac{1}{2}h, \frac{1}{2}(y_n + y_{n+1}) \right).$$

求上述公式的精度, 并给出局部截断误差的首项, 并求其绝对稳定区间.

解. 先写出局部截断误差

$$T = y(x_{n+1}) - \left[y(x_n) + hf \left(x_n + \frac{h}{2}, \frac{y(x_n) + y(x_{n+1})}{2} \right) \right].$$

为书写简便, 记 $y = y(x_n)$, $y' = y'(x_n) = f$, $y'' = y''(x_n)$ 等. 所有的偏导数 f_x, f_y, f_{xx} 等均默认在 (x_n, y_n) 处取值. 先展开简单的

$$y(x_{n+1}) = y + hy' + \frac{h^2}{2}y'' + \frac{h^3}{6}y''' + \mathcal{O}(h^4).$$

下面要进行二元Taylor展开:

$$\begin{aligned} f(x_n + \Delta x, y_n + \Delta y) &= f + f_x \Delta x + f_y \Delta y \\ &\quad + \frac{1}{2} [f_{xx} (\Delta x)^2 + 2f_{xy} \Delta x \Delta y + f_{yy} (\Delta y)^2] + \mathcal{O}(h^3), \end{aligned}$$

这里 $\Delta x = \frac{h}{2}$, 而

$$\begin{aligned} \Delta y &= \frac{y(x_n) + y(x_{n+1})}{2} - y(x_n) = \frac{y(x_{n+1}) - y(x_n)}{2} \\ &= \frac{1}{2} \left(hy' + \frac{h^2}{2}y'' + \mathcal{O}(h^3) \right) = \frac{h}{2}y' + \frac{h^2}{4}y'' + \mathcal{O}(h^3). \end{aligned}$$

故整理得

$$\begin{aligned}f(\cdots) &= f + \frac{h}{2}y'' + h^2 \left[\frac{1}{4}f_y y'' + \frac{1}{8}(f_{xx} + 2f_{xy}f + f_{yy}f^2) \right] + \mathcal{O}(h^3) \\&= f + \frac{h}{2}y'' + h^2 \left[\frac{1}{4}f_y y'' + \frac{1}{8}(y''' - f_y y'') \right] + \mathcal{O}(h^3) \\&= f + \frac{h}{2}y'' + h^2 \left[\frac{1}{8}y''' + \frac{1}{8}f_y y'' \right] + \mathcal{O}(h^3).\end{aligned}$$

故局部截断误差为

$$\begin{aligned}T &= \left(y + hy' + \frac{h^2}{2}y'' + \frac{h^3}{6}y''' \right) \\&\quad - \left(y + hf + \frac{h^2}{2}y'' + \frac{h^3}{8}y''' + \frac{h^3}{8}f_y y'' \right) + \mathcal{O}(h^4) \\&= \left(\frac{1}{6} - \frac{1}{8} \right) h^3 y''' - \frac{1}{8} h^3 f_y y'' + \mathcal{O}(h^4) \\&= \frac{1}{24} h^3 y''' - \frac{1}{8} h^3 f_y y'' + \mathcal{O}(h^4),\end{aligned}$$

故局部截断误差的首项为 $(\frac{1}{24}y'''(x_n) - \frac{1}{8}f_y(x_n, y_n)y''(x_n))h^3$, 具有**2阶**精度.

下面求绝对稳定区间. 考虑 $y' = \lambda y (\lambda < 0)$, 并置 $z := h\lambda$, 代入

$$\begin{aligned}y_{n+1} = y_n + h\lambda \left(\frac{y_n + y_{n+1}}{2} \right) &\iff y_{n+1} = y_n + \frac{z}{2}(y_n + y_{n+1}) \\&\iff y_{n+1} \left(1 - \frac{z}{2} \right) = y_n \left(1 + \frac{z}{2} \right).\end{aligned}$$

令 $R(z) = \frac{1+\frac{z}{2}}{1-\frac{z}{2}}$, 只需让 $|R(z)| < 1$, 解得 $z < 0$, 故绝对稳定区间为 $(-\infty, 0)$. $\square$

注. 由于测试方程中的 f 仅依赖于 y , 所以 $x_n + \frac{1h}{2}$ 这一项不影响计算.

练习

练习1. $y_{n+1} = y_n + \frac{h}{12}[5f_{n+1} + 8f_n - f_{n-1}]$.

解. 局部截断误差为

$$T = y(x_{n+1}) - \left\{ y(x_n) + \frac{h}{12} [5y'(x_{n+1}) + 8y'(x_n) - y'(x_{n-1})] \right\}.$$

记 $y = y(x_n)$, $y' = y'(x_n)$ 等. 先展开 $y(x_{n+1})$:

$$y(x_{n+1}) = y + hy' + \frac{h^2}{2}y'' + \frac{h^3}{6}y''' + \frac{h^4}{24}y^{(4)} + \mathcal{O}(h^5).$$

再展开 $y'(x_{n+1})$ 和 $y'(x_{n-1})$:

$$y'(x_{n+1}) = y' + hy'' + \frac{h^2}{2}y''' + \frac{h^3}{6}y^{(4)} + \mathcal{O}(h^4),$$

$$y'(x_{n-1}) = y' - hy'' + \frac{h^2}{2}y''' - \frac{h^3}{6}y^{(4)} + \mathcal{O}(h^4).$$

整理得

$$\begin{aligned} & 5y'(x_{n+1}) + 8y'(x_n) - y'(x_{n-1}) \\ &= 5 \left(y' + hy'' + \frac{h^2}{2}y''' + \frac{h^3}{6}y^{(4)} \right) + 8y' - \left(y' - hy'' + \frac{h^2}{2}y''' - \frac{h^3}{6}y^{(4)} \right) + \mathcal{O}(h^4) \\ &= 12y' + 6hy'' + 2h^2y''' + h^3y^{(4)} + \mathcal{O}(h^4). \end{aligned}$$

继续:

$$\begin{aligned} \text{大括号里的} &= y + \frac{h}{12} \left[12y' + 6hy'' + 2h^2y''' + h^3y^{(4)} + \mathcal{O}(h^4) \right] \\ &= y + hy' + \frac{6}{12}h^2y'' + \frac{2}{12}h^3y''' + \frac{1}{12}h^4y^{(4)} + \mathcal{O}(h^5) \\ &= y + hy' + \frac{1}{2}h^2y'' + \frac{1}{6}h^3y''' + \frac{1}{12}h^4y^{(4)} + \mathcal{O}(h^5). \end{aligned}$$

故局部截断误差为

$$T = \left(\frac{1}{24} - \frac{1}{12} \right) h^4 y^{(4)} + \mathcal{O}(h^5) = -\frac{1}{24} h^4 y^{(4)}(x_n) + \mathcal{O}(h^5).$$

故局部截断误差的首项为 $-\frac{1}{24}h^4y^{(4)}(x_n)$, 具有**3阶**精度.

下面求绝对稳定区间. 令 $y' = \lambda y$ (其中 $\lambda < 0$ 为实数), 并令 $z = h\lambda$.

代入差分格式:

$$\begin{aligned} y_{n+1} &= y_n + \frac{z}{12}(5y_{n+1} + 8y_n - y_{n-1}) \\ \iff (12 - 5z)y_{n+1} - (12 + 8z)y_n + zy_{n-1} &= 0. \end{aligned}$$

对应的特征方程(设特征根为 r)为:

$$P(r) = (12 - 5z)r^2 - (12 + 8z)r + z = 0.$$

只需确定

$$P(1) > 0, \quad P(-1) > 0, \quad -1 < \frac{6+4z}{12-5z} < 1,$$

解得 $z \in (-6, 0)$. $\square$

练习2. $y_{n+1} = y_n + hf\left(x_n + \frac{h}{2}, y_n + \frac{h}{2}f(x_n, y_n)\right)$.

解. 先写出

$$T = y(x_{n+1}) - \left[y(x_n) + hf\left(x_n + \frac{h}{2}, y(x_n) + \frac{h}{2}f(x_n, y(x_n))\right) \right].$$

选定展开点 x_n , 进行Taylor展开. 为书写简便, 记 $y = y(x_n)$, 以及 f, f_x, f_y, f_{xx} 等均表示在 $(x_n, y(x_n))$ 处的取值. 首先,

$$y(x_{n+1}) = y + hy' + \frac{h^2}{2}y'' + \frac{h^3}{6}y''' + \mathcal{O}(h^4).$$

下面需要展开 $f(x_n + \Delta x, y_n + \Delta y)$, 其中 $\Delta x = \frac{h}{2}, \Delta y = \frac{h}{2}y'$. 利用二元Taylor展开至二次项:

$$\begin{aligned} k_2 &= f\left(x_n + \frac{h}{2}, y + \frac{h}{2}y'\right) \\ &= y' + \left(f_x \frac{h}{2} + f_y \frac{h}{2}y'\right) + \frac{1}{2} \left[f_{xx} \left(\frac{h}{2}\right)^2 + 2f_{xy} \left(\frac{h}{2}\right) \left(\frac{h}{2}y'\right) + f_{yy} \left(\frac{h}{2}y'\right)^2 \right] + \mathcal{O}(h^3) \\ &= y' + \frac{h}{2}(f_x + f_y y') + \frac{h^2}{8}(f_{xx} + 2f_{xy}y' + f_{yy}y'^2) + \mathcal{O}(h^3). \end{aligned}$$

利用全微分关系 $y'' = f_x + f_y y'$ 以及 $y''' - f_y y'' = f_{xx} + 2f_{xy}y' + f_{yy}y'^2$, 整理得

$$k_2 = y' + \frac{h}{2}y'' + \frac{h^2}{8}(y''' - f_y y'') + \mathcal{O}(h^3).$$

代入并按 h 的幂次合并同类项. 将 k_2 的展开式代回公式的右端:

$$\begin{aligned} \text{RHS} &= y + hk_2 \\ &= y + h \left[y' + \frac{h}{2}y'' + \frac{h^2}{8}(y''' - f_y y'') + \mathcal{O}(h^3) \right] \\ &= y + hy' + \frac{h^2}{2}y'' + \frac{h^3}{8}y''' - \frac{h^3}{8}f_y y'' + \mathcal{O}(h^4). \end{aligned}$$

故

$$\begin{aligned}T_{n+1} &= \left(y + hy' + \frac{h^2}{2}y'' + \frac{h^3}{6}y''' \right) - \left(y + hy' + \frac{h^2}{2}y'' + \frac{h^3}{8}y''' - \frac{h^3}{8}f_y y'' \right) + \mathcal{O}(h^4) \\&= \left(\frac{1}{6} - \frac{1}{8} \right) h^3 y''' - \left(-\frac{1}{8} \right) h^3 f_y y'' + \mathcal{O}(h^4) \\&= \frac{1}{24} h^3 y''' + \frac{1}{8} h^3 f_y y'' + \mathcal{O}(h^4) \\&= \left[\frac{1}{24} y'''(x_n) + \frac{1}{8} f_y(x_n, y_n) y''(x_n) \right] h^3 + \mathcal{O}(h^4).\end{aligned}$$

故局部截断误差的首项为 $\left[\frac{1}{24} y'''(x_n) + \frac{1}{8} f_y(x_n, y_n) y''(x_n) \right] h^3$, 具有二阶精度.

设 $y' = \lambda y$, 令 $z = h\lambda$, 则

$$y_{n+1} = y_n + h \cdot \lambda \left(y_n + \frac{h}{2} \lambda y_n \right) = \left(1 + z + \frac{z^2}{2} \right) y_n,$$

令 $R(z) := 1 + z + \frac{z^2}{2}$, 只需考虑 $|R(z)| < 1$. 解得 $-2 < z < 0$, 故绝对稳定区间为 $(-2, 0)$. $\square$

练习3. 已知求解初值问题

$$\begin{cases} y' = f(x, y), & x \in [a, b] \\ y(a) = y_0, \end{cases}$$

的数值公式为

$$y_{n+1} = y_n + hf \left(x_n + \frac{2h}{3}, \frac{y_n + 2y_{n+1}}{3} \right).$$

求该公式的精度, 给出局部截断误差的首项, 并求绝对稳定区间.

解. 先写出局部截断误差

$$T = y(x_{n+1}) - y(x_n) - hf \left(x_n + \frac{2h}{3}, \frac{y(x_n) + 2y(x_{n+1})}{3} \right).$$

为了方便书写, 以下记 $y_n = y(x_n)$, $y'_n = y'(x_n)$, $y''_n = y''(x_n)$ 等, 且 $f = f(x_n, y_n)$. 首先, 有

$$y(x_{n+1}) = y_n + hy'_n + \frac{h^2}{2} y''_n + \mathcal{O}(h^3),$$

记 $\tilde{y} := \frac{y_n + 2y(x_{n+1})}{3}$, 则

$$\begin{aligned}\tilde{y} &= \frac{1}{3}y_n + \frac{2}{3}\left(y_n + hy'_n + \frac{h^2}{2}y''_n + \mathcal{O}(h^3)\right) \\ &= y_n + \frac{2}{3}hy'_n + \frac{1}{3}h^2y''_n + \mathcal{O}(h^3),\end{aligned}$$

故 $\Delta y = \tilde{y} - y_n = \frac{2}{3}hy'_n + \mathcal{O}(h^2)$. 因此,

$$\begin{aligned}f(x_n + \frac{2}{3}h, y_n + \Delta y) &= f + f_x\left(\frac{2}{3}h\right) + f_y\left(\frac{2}{3}hy'_n\right) + \mathcal{O}(h^2) \\ &= f + \frac{2}{3}h(f_x + f_y y'_n) + \mathcal{O}(h^2) \\ &= y'_n + \frac{2}{3}hy''_n + \mathcal{O}(h^2).\end{aligned}$$

故

$$\begin{aligned}T &= \left(y_n + hy'_n + \frac{h^2}{2}y''_n + \mathcal{O}(h^3)\right) - y_n - h\left(y'_n + \frac{2}{3}hy''_n + \mathcal{O}(h^2)\right) \\ &= hy'_n + \frac{1}{2}h^2y''_n - hy'_n - \frac{2}{3}h^2y''_n + \mathcal{O}(h^3) \\ &= \left(\frac{1}{2} - \frac{2}{3}\right)h^2y''_n + \mathcal{O}(h^3) \\ &= -\frac{1}{6}h^2y''_n + \mathcal{O}(h^3),\end{aligned}$$

故**局部截断误差的首项**为 $-\frac{1}{6}h^2y''(x_n)$, 具有一阶精度.

下面求绝对稳定区间. 考虑 $y' = \lambda y (\lambda < 0)$, 并记 $z := h\lambda$, 代入得

$$\begin{aligned}y_{n+1} &= y_n + h\left[\lambda\left(\frac{y_n + 2y_{n+1}}{3}\right)\right] = y_n + z\left(\frac{y_n + 2y_{n+1}}{3}\right) \\ \iff y_{n+1}\left(1 - \frac{2}{3}z\right) &= y_n\left(1 + \frac{z}{3}\right),\end{aligned}$$

故令 $R(z) := \frac{3+z}{3-2z}$, 只需让 $|R(z)| < 1$, 解得 $z < 0$, 故绝对稳定区间为 $(-\infty, 0)$. $\square$